

Hammad Ali Tahir

[Portfolio](#) | [Github](#) | +923334247884 | hammadalitahir8@gmail.com | [Linkedin](#)

SUMMARY

Final-year BS Computer Science student specializing in ML, NLP, and LLM fine-tuning, with hands-on experience building RAG pipelines, deploying full-stack AI applications, and achieving production-grade model accuracy. Seeking an ML Engineer, NLP Engineer, or AI Engineer role to build and deploy production-ready AI systems at scale.

EXPERIENCE

[AI Engineer | AI-Based Legal Case Management System](#)

Uraan AI Techathon 1.0

Sep 2025 - Oct 2025

- Built a production-grade AI LegalTech SaaS platform with intelligent case classification, automated triage, and semantic precedent retrieval using ML, NLP, and RAG pipelines.
- Achieved **91.5%** classification accuracy and **95.2%** prioritization precision using ensemble models (SVM, Naïve Bayes, Random Forest), improving legal data analysis efficiency by **40%+**.
- Developed full-stack AI application (React.js, Python/Streamlit) integrating Hugging Face transformer embeddings, Qdrant vector database, and Llama-3.1-8B LLM for context-aware legal research and generative AI insights.
- Deployed cloud-native solution on Render and Vercel with CI/CD pipelines, RESTful APIs, and vector similarity search for scalable Legal AI delivery.

[AI Developer & Consultant | Academic Chatbot System](#)

University of Education Lahore

Jan 2026 - Present

- Architected and deployed a full-stack AI-powered academic chatbot serving the University of Education, Lahore, logging **100+ production queries** via LangSmith tracing with documented **user satisfaction scores of 1.0** across multiple interactions, implementing a Retrieval-Augmented Generation (RAG) pipeline with Pinecone vector search, OpenAI GPT-4o-mini, and a self-correcting Smart RAG loop (query rewriting, chunk grading, multi-retry retrieval) across 3 institutional knowledge bases covering BS/ADP, MS/PhD, and university regulations.
- Developed a production-grade React + Vite PWA frontend with real-time Server-Sent Events (SSE) streaming, Zustand state management, responsive mobile-first design, and configurable pipeline controls (query enhancement, reranking, Smart RAG), delivering a ChatGPT-like conversational UX with source attribution and relevance scoring.
- Engineered a scalable FastAPI backend integrating Pinecone vector database, HuggingFace cross-encoder reranking, Redis Cloud session-aware conversation memory, and OpenAI embedding and generation models via a modular pipeline architecture supporting both streaming and non-streaming inference for low-latency academic Q&A.

PROJECTS

[Loan Approval Prediction System](#)

Python | Flask | Scikit-learn | Docker | Render

Jun 2025 - Jul 2025

- Built an end-to-end Loan Approval Prediction System using Python, Flask, scikit-learn, and Gunicorn, implementing a complete ML pipeline, data ingestion, feature engineering (StandardScaler, OneHotEncoder, ColumnTransformer), and automated model selection across 6 classifiers (Gradient Boosting, Random Forest, AdaBoost, SVC, Logistic Regression, Decision Tree) achieving 97.6% accuracy with **0.99 ROC-AUC**.
- Deployed a production-ready ML web application on Render with Docker support, featuring RESTful prediction APIs, modular training/prediction pipelines, health monitoring, structured logging, and custom exception handling, following industry-standard CI/CD and scalable software engineering practices.

- Built an AI-powered GitHub research agent using LangGraph and FastAPI, implementing an automated multi-step pipeline that retrieves repository metadata (stars, forks, languages, activity), analyzes architecture, inspects issues, and synthesizes caching-enabled research reports via a ReAct agent equipped with live GitHub tools.
- Developed a full-stack application with a React + Vite frontend deployed on Vercel and a Python/FastAPI backend deployed on Render, featuring a responsive real-time chat interface for interactive follow-up Q&A powered by a LangGraph ReAct reasoning loop.

EDUCATION

BS Computer Science-University of Education Lahore

CGPA: 3.21/ 4.0 | Expected Graduation: June 2026

- *Relevant Coursework: Linear Algebra, Statistics, Calculus-I, OOP, DSA, Machine Learning, Data Science, Database Systems*

SKILLS

Languages: Python, SQL

ML/AI Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost

LLM/NLP: Hugging Face, LangChain, LangGraph, OpenAI API, RAG, Fine-tuning, NLTK

Vector Databases: Pinecone, Qdrant, ChromaDB

Backend: FastAPI, Flask, Streamlit, RESTful APIs, Redis

DevOps/Cloud: Docker, Git, Render, Vercel, CI/CD

Data: Pandas, NumPy, EDA, Feature Engineering

Concepts: CNNs, NLP, Supervised/Unsupervised Learning, Ensemble Methods, Vector Similarity Search

Soft Skills: Adaptable, fast learner, strong communicator, collaborative team player

CERTIFICATION

- **Supervised Machine Learning: Regression and Classification** — [DeepLearning.AI](#) & Stanford University | 2025
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — Oracle University | Aug 2025